

Human review packet: GKGM19IterativeLocalVoting

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: GKGM19IterativeLocalVoting
Paper id: GKGM19IterativeLocalVoting
Source version: not recorded
Source record: audit/paper_statement_map.json
Rows: 44
Generated: 2026-08-18
Source URL: [official source](#)

Review status: 0/44 source-claim screens; 0/8 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper GKGM19IterativeLocalVoting --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

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Material library prerequisites

EconCSLib.FiniteDimensionalNorms.coordinateLinearFunctional

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → [inst : Fintype ι] → (g : ι → ℝ) → ∑ i, g i • ContinuousLinearMap.proj i
```

Exact Lean library declaration

```
def coordinateLinearFunctional {ι : Type*} [Fintype ι]
  (g : ι → ℝ) : (ι → ℝ) → L[ℝ] ℝ :=
  ∑ i : ι, g i •
  (ContinuousLinearMap.proj (R := ℝ) (φ := fun _ : ι => ℝ) i :
    (ι → ℝ) → L[ℝ] ℝ)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.FiniteDimensionalNorms.l1

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → [inst : Fintype ι] → (x : ι → ℝ) → ∑ i, |x i|
```

Exact Lean library declaration

```
def l1 {ι : Type*} [Fintype ι] (x : ι → ℝ) : ℝ :=  
  ∑ i : ι, |x i|
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.FiniteDimensionalNorms.l2Sq

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → [inst : Fintype ι] → (x : ι → ℝ) → ∑ i, x i ^ 2
```

Exact Lean library declaration

```
def l2Sq {ι : Type*} [Fintype ι] (x : ι → ℝ) : ℝ :=  
  ∑ i : ι, x i ^ 2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.FiniteDimensionalNorms.l2

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → [inst : Fintype ι] → (x : ι → ℝ) → √(EconCSLib.FiniteDimensionalNorms.l2Sq x)
```

Exact Lean library declaration

```
def l2 {ι : Type*} [Fintype ι] (x : ι → ℝ) : ℝ :=  
  Real.sqrt (l2Sq x)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.FiniteDimensionalNorms.linf

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → [inst : Fintype ι] → [inst_1 : Nonempty ι] → (x : ι → ℝ) → Finset.univ.sup' - fun i => |x i|
```

Exact Lean library declaration

```
def linf {ι : Type*} [Fintype ι] [Nonempty ι] (x : ι → ℝ) : ℝ :=  
  (Finset.univ : Finset ι).sup' Finset.univ_nonempty (fun i => |x i|)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.FiniteDimensionalNorms.LpPower

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\iota : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype } \iota] \rightarrow (p : \mathbb{R}) \rightarrow (x : \iota \rightarrow \mathbb{R}) \rightarrow \sum i, |x\ i|^\wedge p$

Exact Lean library declaration

```
def lpPower {ι : Type*} [Fintype ι] (p : ℝ) (x : ι → ℝ) : ℝ :=  
  ∑ i : ι, |x i| ^ p
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.FiniteDimensionalNorms.Lp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} → [inst : Fintype ι] → (p : ℝ) → (x : ι → ℝ) → EconCSLib.FiniteDimensionalNorms.LpPower p x ^ (1 / p)
```

Exact Lean library declaration

```
def lp {ι : Type*} [Fintype ι] (p : ℝ) (x : ι → ℝ) : ℝ :=  
  (lpPower p x) ^ (1 / p)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.HasBoundedDensity

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall \{\alpha : \text{Type } u_1\} [\text{inst} : \text{MeasurableSpace } \alpha] (\nu \mu : \text{MeasureTheory.Measure } \alpha) (C : \text{ENNReal}),$
 $\exists D, \mu = \nu.\text{withDensity } D \wedge \forall^m (x : \alpha) \partial \nu, D \ x \leq C$

Exact Lean library declaration

```
def HasBoundedDensity
  {α : Type*} [MeasurableSpace α]
  (ν μ : Measure α) (C : ℝ≥0∞) : Prop :=
  ∃ D : α → ℝ≥0∞, μ = ν.withDensity D ∧ ∀m x ∂ν, D x ≤ C
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:
Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$,
termination time T , norm q for local neighborhood.
Output:
Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$
and elicit

x'
 $t = \arg \max_{x \in \{s : \|s - x_{t-1}\|_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [x'$
 $t]_X$, where $[\cdot]_X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{l, m \in \{t-N, \dots, t\}} \|x_l - x_m\| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall \{ \text{Point} : \text{Type } u_1 \} (G : \text{GKGMM19IterativeLocalVoting.ILVGeometryFormulaData Point})$
 $(q : \text{GKGMM19IterativeLocalVoting.SourceNorm}) (\text{center candidate} : \text{Point}) (r : \mathbb{R}),$
 $\text{candidate} \in \text{GKGMM19IterativeLocalVoting.localNeighborhoodFormulaData } G \text{ } q \text{ center } r \leftrightarrow$
 $\text{candidate} \in G.\text{solutionSpace} \wedge G.\text{normDistance } q \text{ candidate center} \leq r$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_local_neighborhood_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:

Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$, termination time T , norm q for local neighborhood.

Output:

Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$ and elicit

x'
 $t = \arg \max_{x \in \{s : ||s - x_t - 1||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [x'$
 $t]X$, where $[\cdot]X$ is a projection onto space X ; i.e. ask the voter to move to her favorite point within constraints, and then project the reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when $\max_{t \in \{t-N, \dots, t\}} |x_t - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

```
∀ {Point : Type u_1} (G : GKGMM19IterativeLocalVoting.ILVGeometryFormulaData Point)
  (q : GKGMM19IterativeLocalVoting.SourceNorm) (project : Point → Point),
  GKGMM19IterativeLocalVoting.isNormProjectionOntoFormulaData G q project →
  ∀ (y : Point), project y ∈ G.solutionSpace ∧ IsMinOn (fun x => G.normDistance q x y) G.solutionSpace (project y)
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_norm_projection_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:

Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$, termination time T , norm q for local neighborhood.

Output:

Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$ and elicit

x'
 $t = \arg \max_{x \in \{s : ||s - x_t - 1||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [x'$
 $t]X$, where $[\cdot]X$ is a projection onto space X ; i.e. ask the voter to move to her favorite point within constraints, and then project the reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when $\max_{x_l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

```
∀ {Point : Type u_1} {G : GKGMM19IterativeLocalVoting.ILVGeometryFormulaData Point}
  {q : GKGMM19IterativeLocalVoting.SourceNorm} {project : Point → Point} {raw trajectory : ℕ → Point},
  GKGMM19IterativeLocalVoting.isNormProjectionOntoFormulaData G q project →
  trajectory 0 ∈ G.solutionSpace →
  (∀ (t : ℕ), GKGMM19IterativeLocalVoting.Algorithm1ProjectedUpdate project (raw t) (trajectory (t + 1))) →
  ∀ (t : ℕ), trajectory t ∈ G.solutionSpace
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_projected_trajectory_feasible_of_normProjection

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:
Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$,
termination time T , norm q for local neighborhood.
Output:
Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$
and elicit

x'
 $t = \arg \max_{x \in \{s: ||s - x_{t-1}||_q \leq r_t\}} f_{v_t}(x),$
(1)

and then compute $x_t = [x'$
 $t]X$, where $[\cdot]X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall \{ \text{Point} : \text{Type } u_1 \} (\text{project} : \text{Point} \rightarrow \text{Point}) (\text{raw next} : \text{Point}),$
GKGMM19IterativeLocalVoting.Algorithm1ProjectedUpdate project raw next $\leftrightarrow$ next = project raw

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_projected_update_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:
Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$,
termination time T , norm q for local neighborhood.
Output:
Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$
and elicit

x'
 $t = \arg \max_{x \in \{s: ||s - x_{t-1}||_q \leq r_t\}} f_{v_t}(x),$
(1)

and then compute $x_t = [x'$
 $t]_X$, where $[\cdot]_X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{x_l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall (r_0 : \mathbb{R}) (t : \mathbb{N}), \text{GKGMM19IterativeLocalVoting.ilvRadius } r_0 \ t = r_0 / t$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_radius_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:

Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$, termination time T , norm q for local neighborhood.

Output:

Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$ and elicit

x'
 $t = \arg \max_{x \in \{s: ||s - x_t - 1||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [\mathbf{x}'$
 $\mathbf{t}]_X$, where $[\cdot]_X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{x_l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall \{r_0 : \mathbb{R}\}, 0 < r_0 \rightarrow \neg \text{Summable } \text{fun } t \Rightarrow \text{GKGMM19IterativeLocalVoting.ilvRadius } r_0 (t + 1)$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_radius_not_summable

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:

Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$, termination time T , norm q for local neighborhood.

Output:

Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$ and elicit

x'
 $t = \arg \max_{x \in \{s: ||s - x_{t-1}||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [x'$
 $t]_X$, where $[\cdot]_X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall (r_0 : \mathbb{R}), \text{Summable } \text{fun } t \Rightarrow \text{GKGMM19IterativeLocalVoting.ilvRadius } r_0 (t + 1) ^ 2$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_radius_sq_summable

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:
Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$,
termination time T , norm q for local neighborhood.
Output:
Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$
and elicit

x'
 $t = \arg \max_{x \in \{s: ||s - x_t - 1||_q \leq r_t\}} f_{v_t}(x),$
(1)

and then compute $x_t = [x'$
 $t]X$, where $[\cdot]X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{x_l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall \{r_0 : \mathbb{R}\}, 0 < r_0 \rightarrow \text{GKGMM19IterativeLocalVoting.SSGMStepSizeConditions } (\text{GKGMM19IterativeLocalVoting.ilvRadius } r_0)$

Lean proof endpoint (built separately)

`GKGMM19IterativeLocalVoting.algorithm1_radius_ssgm_step_size_conditions`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:
Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$,
termination time T , norm q for local neighborhood.

Output:
Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$
and elicit

x'
 $t = \arg \max_{x \in \{s: ||s - x_{t-1}||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [x'$
 $t]X$, where $[\cdot]X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

```
∀ {r0 : ℝ},  
  0 < r0 →  
    Filter.Tendsto (fun n => ∑ t ∈ Finset.range n, GKGMM19IterativeLocalVoting.ilvRadius r0 (t + 1)) Filter.atTop  
      Filter.atTop
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_radius_sum_tendsto_atTop

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:

Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$, termination time T , norm q for local neighborhood.

Output:

Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$ and elicit

x'
 $t = \arg \max_{x \in \{s: ||s - x_t - 1||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [x'$
 $t]X$, where $[\cdot]X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{x_l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall (r_0 : \mathbb{R}), \text{Filter.Tendsto } (\text{GKGMM19IterativeLocalVoting.ilvRadius } r_0) \text{ Filter.atTop } (\text{nhds } 0)$

Lean proof endpoint (built separately)

`GKGMM19IterativeLocalVoting.algorithm1_radius_tendsto_zero`

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:
Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$,
termination time T , norm q for local neighborhood.
Output:
Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$
and elicit

x'
 $t = \arg \max_{x \in \{s : ||s - x_t - 1||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [x'$
 $t]X$, where $[\cdot]X$ is a projection onto space X ; i.e. ask
the voter to move to her favorite point within constraints, and then project the
reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when
 $\max_{l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

$\forall \{ \text{Point} : \text{Type } u_1 \} (G : \text{GKGMM19IterativeLocalVoting.ILVGeometryFormulaData Point})$
 $(q : \text{GKGMM19IterativeLocalVoting.SourceNorm}) (\text{trajectory} : \mathbb{N} \rightarrow \text{Point}) (T \ N \ t : \mathbb{N}) (\text{epsilon} : \mathbb{R}),$
 $\text{GKGMM19IterativeLocalVoting.algorithm1StopConditionFormulaData } G \ q \ \text{trajectory } T \ N \ t \ \text{epsilon} \leftrightarrow$
 $t = T \vee \text{GKGMM19IterativeLocalVoting.algorithm1WindowStableFormulaData } G \ q \ \text{trajectory } t \ N \ \text{epsilon}$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_stop_condition_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 170-192

Verbatim source input

Algorithm 1: Iterative Local Voting (ILV)

Inputs:

Initial solution $x_0 \in X$, tolerance $\epsilon > 0$, an integer N , initial radius $r_0 > 0$, termination time T , norm q for local neighborhood.

Output:

Solution x .

• For $t \geq 1$, sample a voter $v_t \in V$ at random from the population; set $r_t = r_0/t$ and elicit

x'
 $t = \arg \max_{x \in \{s : ||s - x_t - 1||_q \leq r_t\}} f_{v_t}(x)$,
(1)

and then compute $x_t = [\mathbf{x}'_t]_X$, where $[\cdot]_X$ is a projection onto space X ; i.e. ask the voter to move to her favorite point within constraints, and then project the reported point onto X .

• Stop when either $t = T$, in which case return x_T , or when $\max_{l, m \in \{t-N, \dots, t\}} |x_l - x_m| \leq \epsilon$, in which case return $x = x_t$.

Expanded PaperInterface specification

```
∀ {Point : Type u_1} (G : GKGMM19IterativeLocalVoting.ILVGeometryFormulaData Point)
  (q : GKGMM19IterativeLocalVoting.SourceNorm) (trajectory : ℕ → Point) (t N : ℕ) (epsilon : ℝ),
  GKGMM19IterativeLocalVoting.algorithm1WindowStableFormulaData G q trajectory t N epsilon ↔
    ∀ (l m : ℕ),
      l ∈ Finset.Icc (t - N) t → m ∈ Finset.Icc (t - N) t → G.normDistance q (trajectory l) (trajectory m) ≤ epsilon
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.algorithm1_window_stable_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 2138-2155

Verbatim source input

Lemma 3. $\forall (p, q) \text{ s.t. } p > 0, q > 0, \text{ and } 1/p + 1/q = 1, \|\nabla\|_p \|x - xv\|_q = 1, \forall x \text{ s.t. } x_m \neq x_m$
v for any m.

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q

neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

Proof. Since the probability of picking a voter v such that $x_m = x_m$
v for some dimension m
is 0, we have $\nabla g_v(x_t) = g_v$ for $g_v = \nabla f_v(x_t)$. Thus we obtain the gradient exactly, and hence Theorem 5 applies with $b_t = 0$ for all t.

C.5 Proof of Propositions

Expanded PaperInterface specification

```
 $\forall \{ \text{Coord} : \text{Type } u_1 \} [\text{inst} : \text{Fintype Coord}]$   
(D : GKGM19IterativeLocalVoting.FiniteCoordinateIdealDistributionData Coord) (x : Coord  $\rightarrow \mathbb{R}$ ),  
 $\forall^m (\text{ideal} : \text{Coord} \rightarrow \mathbb{R}) \partial D.\text{idealMeasure}, \forall (i : \text{Coord}), x \ i \neq \text{ideal } i$ 
```

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.c3_product_density_coordinate_noncollision_ae

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 410-420

Verbatim source input

behavior models. For the rest of the technical analysis, we make the following assumptions on our model.

C1 The solution space $X \subseteq \mathcal{RM}$ is non-empty, bounded, closed, and convex.

C2 Each voter v has a unique ideal solution $x_v \in X$.

C3 The ideal point x_v of each voter is drawn independently from a probability distribution with a bounded and measurable density function h_X .

Under this model, for a solution $x \in X$, the societal utility is given by $E_v[f_v(x)]$. and the

Expanded PaperInterface specification

```
∀
  (solutionSpace_nonempty_bounded_closed_convex uniqueIdealSolutions idealDistribution_bounded_measurable_density :
    Prop),
  solutionSpace_nonempty_bounded_closed_convex ∧ uniqueIdealSolutions ∧ idealDistribution_bounded_measurable_density ↔
    solutionSpace_nonempty_bounded_closed_convex ∧ uniqueIdealSolutions ∧ idealDistribution_bounded_measurable_density
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.conditions_c123_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. L_p normed utilities. The voter utility function is L_p normed if $f_v(x) = -||x - x_v||_p$, $\forall x \in X$.

Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2)$, $(1, \infty)$, or $(\infty, 1)$.

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Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Coord : Type u_2} [inst : Fintype Coord] [inst_1 : Nonempty Coord]
  (utility : Voter → (Coord → ℝ) → ℝ) (ideal : Voter → Coord → ℝ),
  (∀ (v : Voter) (x : Coord → ℝ),
    utility v x =
      -GKGMM19IterativeLocalVoting.finiteCoordinateDistance GKGMM19IterativeLocalVoting.SourceNorm.l1 x (ideal v)) ↔
  ∀ (v : Voter) (x : Coord → ℝ), utility v x = -EconCSLib.FiniteDimensionalNorms.l1 fun m => x m - ideal v m
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.definition1_finite_coordinate_l1_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. L_p normed utilities. The voter utility function is L_p normed if $f_v(x) = -||x - x_v||_p$, $\forall x \in X$.

Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2)$, $(1, \infty)$, or $(\infty, 1)$.

322

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Coord : Type u_2} [inst : Fintype Coord] [inst_1 : Nonempty Coord]
  (utility : Voter → (Coord → ℝ) → ℝ) (ideal : Voter → Coord → ℝ),
  (∀ (v : Voter) (x : Coord → ℝ),
    utility v x =
      -GKGMM19IterativeLocalVoting.finiteCoordinateDistance GKGMM19IterativeLocalVoting.SourceNorm.l2 x (ideal v)) ↔
  ∀ (v : Voter) (x : Coord → ℝ), utility v x = -EconCSLib.FiniteDimensionalNorms.l2 fun m => x m - ideal v m
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.definition1_finite_coordinate_l2_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. L_p normed utilities. The voter utility function is L_p normed if $fv(x) = -||x - xv||_p$, $\forall x \in X$.

Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2)$, $(1, \infty)$, or $(\infty, 1)$.

322

Expanded PaperInterface specification

```

V {Voter : Type u_1} {Coord : Type u_2} [inst : Fintype Coord] [inst_1 : Nonempty Coord]
  (utility : Voter → (Coord → ℝ) → ℝ) (ideal : Voter → Coord → ℝ),
  (∀ (v : Voter) (x : Coord → ℝ),
    utility v x =
      -GKMMM19IterativeLocalVoting.finiteCoordinateDistance GKMMM19IterativeLocalVoting.SourceNorm.linf x
        (ideal v)) ↔
    ∀ (v : Voter) (x : Coord → ℝ), utility v x = -EconCSLib.FiniteDimensionalNorms.linf fun m => x m - ideal v m

```

Lean proof endpoint (built separately)

GKMMM19IterativeLocalVoting.definition1_finite_coordinate_linf_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. L_p normed utilities. The voter utility function is L_p normed if $fv(x) = -||x - xv||_p$, $\forall x \in X$.

Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2)$, $(1, \infty)$, or $(\infty, 1)$.

322

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Coord : Type u_2} [inst : Fintype Coord] [inst_1 : Nonempty Coord]
  (utility : Voter → (Coord → ℝ) → ℝ) (ideal : Voter → Coord → ℝ) (p : ℝ),
  (∀ (v : Voter) (x : Coord → ℝ),
    utility v x =
      -GKGMM19IterativeLocalVoting.finiteCoordinateDistance (GKGMM19IterativeLocalVoting.SourceNorm.lp p) x
      (ideal v)) ↔
  ∀ (v : Voter) (x : Coord → ℝ), utility v x = -EconCSLib.FiniteDimensionalNorms.lp p fun m => x m - ideal v m
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.definition1_finite_coordinate_lp_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 19

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. Lp normed utilities. The voter utility function is Lp normed if $fv(x) = -||x - xv||_p$, $\forall x \in X$.

Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are Lp normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with Lq neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2)$, $(1, \infty)$, or $(\infty, 1)$.

322

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Point : Type u_2} (U : GKGMM19IterativeLocalVoting.ILVUtilityFormulaData Voter Point)
  (ideal : Voter → Point) (p : GKGMM19IterativeLocalVoting.SourceNorm),
  GKGMM19IterativeLocalVoting.isLpNormedUtilitiesFormulaData U ideal p ↔
  ∀ (v : Voter) (x : Point), U.utility v x = -U.normDistance p x (ideal v)
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.definition1_lp_normed_utilities_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 20

Source locator: cited publication, lines 480-535

Verbatim source input

sampling such a voter decreases fast enough).

Next, we introduce another general class of utility functions, which we call Weighted Euclidean utilities, for which one can obtain convergence to a unique solution.

Definition 2. Weighted Euclidean utilities. Let the solution space X be decomposable into K disjoint sub-spaces, so that $x = (x_1, \dots, x_K)$ for each $x \in X$ (where $\dim(x_k) = k$). Suppose that the utility function of the voter v is

$f_v(x) = -$

$\sum_{k=1}^K$

w_k

v

$\|w_v\|^2$
 $\|x_k - x_k\|^2$
 $v\|^2$.

where w_v is a voter-specific weight vector, then the function is a Weighted Euclidean utility function. We further assume that $w_v \in \mathbb{R}^K$ and x_v are independently drawn for each voter v from a joint probability distribution with a bounded and measurable density function, with W nonempty, bounded, closed, and convex.

This utility function can be interpreted as follows: the decision-making problem is decomposable into K sub-problems, and each voter v has an ideal point x_k and a weight w_k

for each sub-problem k , so that the voter's disutility for a solution is the weighted sum of the Euclidean distances to the ideal points in each sub-problems. Such utility functions may emerge in facility location problems, for example, where voters have preferences on

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[form-feed in source extraction]

Garg, Kamble, Goel, Marn, & Munagala

the locations of multiple facilities on a map. This utility form is also the one most closely related to the existing literature on Directional Equilibria and Quadratic Voting, in which preferences are linear. To recover the weighted linear preferences case, set $K = M$, with each sub-space of dimension 1. In this case, the following holds:

Proposition 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are Weighted Euclidean, and voters correctly respond to query (1) according to either Model A or Model B. Then, ILV with L2 neighborhoods converges with probability 1 to the societal optimal point.

The intuition for the result is as follows: as long as the neighborhood does not contain the ideal point of the sampled voter, the correct response to query (1) under weighted Euclidean preferences is to move the solution in the direction of the ideal point to the neighborhood boundary, which, as it turns out, is the same as the direction of the gradient. Thus with radius r_t , the effective movement is

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Point : Type u_2} {Component : Type u_3} (utility : Voter → Point → ℝ)
(W : GKGM19IterativeLocalVoting.WeightedEuclideanStructure Voter Point Component),
(W.weightsAndIdealsDistributionCondition ∧
  ∀ (v : Voter) (x : Point),
    utility v x = -∑ k ∈ W.components, W.weight v k / W.weightNorm2 v * W.componentDistance k x v) ↔
(W.weightsAndIdealsDistributionCondition ∧
  ∀ (v : Voter) (x : Point),
    utility v x = -∑ k ∈ W.components, W.weight v k / W.weightNorm2 v * W.componentDistance k x v)
```

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.definition2_weighted_euclidean_utilities_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 21

Source locator: cited publication, lines 540-575

Verbatim source input

normalized (i.e. not divided by $\|w\|_2$), the algorithm would converge to the same point, as if utility functions were normalized.

3.2 Decomposable Utilities

Next consider the general class of decomposable utilities, motivated by the fact that the algorithm with L_∞ neighborhoods is of special interest since they are easy for humans to understand: one can change each dimension up to a certain amount, independent of the others.

Definition 3. Decomposable utilities. A voter utility function is decomposable if there exists concave functions f_m

v
for $m \in \{1 \dots M\}$ such that $f_v(x) = \sum_{m=1}^M f_m(x_m)$.

If the utility functions for the voters are decomposable, then we can show that our algorithm under L_∞ neighborhoods converges to the vector of medians of voters' ideal points on each dimension. Suppose that h_m
 X is the marginal density function of the random variable x_m
 v , and let $\tilde{x}_m$ be the set of medians of x_m
 v . (By set of medians, we mean the set of points such that, on each dimension, the mass of voters with ideal points above and below.)

Proposition 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are decomposable, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_∞ neighborhoods converges with probability 1 to a point in the set of medians $\tilde{x}$.

Although simply eliciting each agent's optimal solution and computing the vector of median allocations on each dimension is a viable approach in the case of decomposable utilities, deciding an optimal allocation across multiple dimensions is a more challenging cognitive task than deciding whether one wants to increase or decrease each dimension relative to the current solution (see Section 5.2.3 for experimental evidence). In fact, in this

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Point : Type u_2} {Coord : Type u_3} (utility : Voter → Point → ℝ)
(D : GKGM19IterativeLocalVoting.DecomposableStructure Voter Point Coord),
(D.coordinateUtilitiesConcave ∧
  ∀ (v : Voter) (x : Point), utility v x = ∑ m ∈ D.coords, D.coordinateUtility m v (D.coordinate m x)) ↔
(D.coordinateUtilitiesConcave ∧
  ∀ (v : Voter) (x : Point), utility v x = ∑ m ∈ D.coords, D.coordinateUtility m v (D.coordinate m x))
```

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.definition3_decomposable_utilities_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 22

Source locator: cited publication, lines 2138-2155

Verbatim source input

Lemma 3. $\forall (p, q) \text{ s.t. } p > 0, q > 0, \text{ and } 1/p + 1/q = 1, \|\nabla\|_x - x v \|_p \|_q = 1, \forall x \text{ s.t.}$
 $x_m \neq x_m$
 v for any m .

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q

neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

Proof. Since the probability of picking a voter v such that x_m
 $t = x_m$
 v for some dimension m
is 0, we have $\nabla g_v t(x_t) = g_v t$ for $g_v t = \nabla f_v t(x_t)$. Thus we obtain the gradient exactly, and hence Theorem 5 applies with $b_t = 0$ for all t .

C.5 Proof of Propositions

Expanded PaperInterface specification

```
∀ {Coord : Type u_1} [Fintype Coord] [inst : MeasurableSpace (Coord → ℝ)] {ν μ : MeasureTheory.Measure (Coord → ℝ)}  
{C : ENNReal},  
EconCSLib.Probability.HasBoundedDensity ν μ C →  
  ∀ (x : Coord → ℝ),  
    (∀ (i : Coord), ν (GKGMM19IterativeLocalVoting.coordinateEqualityHyperplane x i) = 0) →  
      μ (GKGMM19IterativeLocalVoting.coordinateEqualityBadEvent x) = 0
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.lemma3_coordinate_equality_bad_event_null_from_boundedDensity

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 23

Source locator: cited publication, lines 2138-2155

Verbatim source input

Lemma 3. $\forall (p, q) \text{ s.t. } p > 0, q > 0, \text{ and } 1/p + 1/q = 1, \|\nabla\|_p \|x - xv\|_q = 1, \forall x \text{ s.t. } x_m \neq x_m$
v for any m.

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q

neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

Proof. Since the probability of picking a voter v such that $x_m = x_m$
v for some dimension m
is 0, we have $\nabla g_v(t) = g_v$ for $g_v = \nabla f_v(t)$. Thus we obtain the gradient exactly, and hence Theorem 5 applies with $b_t = 0$ for all t.

C.5 Proof of Propositions

Expanded PaperInterface specification

```
∀ {Coord : Type u_1} [inst : Fintype Coord] (p : MeasureTheory.Measure ℝ) [MeasureTheory.SigmaFinite p]
  [MeasureTheory.NoAtoms p] {μ : MeasureTheory.Measure (Coord → ℝ)} {C : ENNReal},
  EconCSLib.Probability.HasBoundedDensity (MeasureTheory.Measure.pi fun x => p) μ C →
    ∀ (x : Coord → ℝ), μ (GKGMM19IterativeLocalVoting.coordinateEqualityBadEvent x) = 0
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.lemma3_coordinate_equality_bad_event_null_from_productMeasure

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 24

Source locator: cited publication, lines 2138-2155

Verbatim source input

Lemma 3. $\forall (p, q) \text{ s.t. } p > 0, q > 0, \text{ and } 1/p + 1/q = 1, ||\nabla||_x -xv||_p||_q = 1, \forall x \text{ s.t.}$
 $x_m \neq x_m$
 v for any m .

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

Proof. Since the probability of picking a voter v such that x_m
 $t = x_m$
 v for some dimension m
is 0, we have $\nabla g_v(t) = g_v$ for $g_v = \nabla f_v(t)$. Thus we obtain the gradient exactly, and hence Theorem 5 applies with $b_t = 0$ for all t .

C.5 Proof of Propositions

Expanded PaperInterface specification

$\forall \{ \text{Coord} : \text{Type } u_1 \} [\text{inst} : \text{Fintype Coord}] (p : \text{MeasureTheory.Measure } \mathbb{R}) [\text{MeasureTheory.SigmaFinite } p]$
 $[\text{MeasureTheory.NoAtoms } p] \{ \mu : \text{MeasureTheory.Measure } (\text{Coord} \rightarrow \mathbb{R}) \} \{ C : \text{ENNReal} \},$
 $\text{EconCSLib.Probability.HasBoundedDensity } (\text{MeasureTheory.Measure.pi fun } x \Rightarrow p) \mu C \rightarrow$
 $\forall (x : \text{Coord} \rightarrow \mathbb{R}), \forall^m (\text{ideal} : \text{Coord} \rightarrow \mathbb{R}) \partial \mu, \forall (i : \text{Coord}), x \ i \neq \text{ideal } i$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.lemma3_coordinate_noncollision_ae_from_productMeasure

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 25

Source locator: cited publication, lines 2138-2155

Verbatim source input

Lemma 3. $\forall (p, q) \text{ s.t. } p > 0, q > 0, \text{ and } 1/p + 1/q = 1, \|\nabla\|_x - x v \|_p \|_q = 1, \forall x \text{ s.t.}$
 $x_m \neq x_m$
 v for any m .

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q

neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

Proof. Since the probability of picking a voter v such that x_m
 $t = x_m$
 v for some dimension m
is 0, we have $\nabla g_v(t) = g_v$ for $g_v = \nabla f_v(t)$. Thus we obtain the gradient exactly, and hence Theorem 5 applies with $b_t = 0$ for all t .

C.5 Proof of Propositions

Expanded PaperInterface specification

```

∀ {Coord : Type u_1} [inst : Fintype Coord] [inst_1 : Nonempty Coord] {p q : ℝ},
  GKGMM19IterativeLocalVoting.HolderDualFinite p q →
  ∀ {x ideal : Coord → ℝ},
    (∀ (i : Coord), x i ≠ ideal i) →
      GKGMM19IterativeLocalVoting.finiteCoordinateNorm (GKGMM19IterativeLocalVoting.SourceNorm.l p q)
        (GKGMM19IterativeLocalVoting.lpCostGradientCandidate p fun i => x i - ideal i) =
1

```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.lemma3_finite_holder_dual_gradient_candidate_norm_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 26

Source locator: cited publication, lines 2138-2155

Verbatim source input

Lemma 3. $\forall(p, q) \text{ s.t. } p > 0, q > 0, \text{ and } 1/p + 1/q = 1, ||\nabla||_x - x v ||_p ||_q = 1, \forall x \text{ s.t. } x_m \neq x_m$
v for any m.

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q

neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

Proof. Since the probability of picking a voter v such that $x_m = x_m$ for some dimension m is 0, we have $\nabla g_v t(x_t) = g_v t$ for $g_v t = \nabla f_v t(x_t)$. Thus we obtain the gradient exactly, and hence Theorem 5 applies with $b_t = 0$ for all t.

C.5 Proof of Propositions

Expanded PaperInterface specification

```
∀ {Coord : Type u_1} [inst : Fintype Coord] [Nonempty Coord] {p : ℝ},
  1 < p →
  ∀ {x ideal : Coord → ℝ},
    (∀ (i : Coord), x i ≠ ideal i) →
      HasFDerivAt (fun y => EconCSLib.FiniteDimensionalNorms.lp p fun i => y i - ideal i)
        (EconCSLib.FiniteDimensionalNorms.coordinateLinearFunctional
          (GKGMM19IterativeLocalVoting.lpCostGradientCandidate p fun i => x i - ideal i))
    x
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.lemma3_gradient_candidate_hasFDerivAt_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 27

Source locator: cited publication, lines 2138-2155

Verbatim source input

Lemma 3. $\forall (p, q) \text{ s.t. } p > 0, q > 0, \text{ and } 1/p + 1/q = 1, \|\nabla\|_x -xv\|_p\|q = 1, \forall x \text{ s.t.}$
 $x_m \neq x_m$
 v for any m .

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q

neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

Proof. Since the probability of picking a voter v such that x_m
 $t = x_m$
 v for some dimension m
is 0, we have $\nabla g_v(t) = g_v$ for $g_v = \nabla f_v(t)$. Thus we obtain the gradient exactly, and hence Theorem 5 applies with $b_t = 0$ for all t .

C.5 Proof of Propositions

Expanded PaperInterface specification

```
∀ {Coord : Type u_1} [inst : Fintype Coord] {p : ℝ},
  0 < p →
  ∀ (d : Coord → ℝ),
    GKGM19IterativeLocalVoting.lpCostGradientCandidate p d = fun i =>
      |d i| ^ (p - 1) * (d i / |d i|) / EconCSLib.FiniteDimensionalNorms.lp p d ^ (p - 1)
```

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.lemma3_gradient_candidate_source_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 28

Source locator: cited publication, lines 118-145

Verbatim source input

bounds on the sum of absolute values of the changes, the sum of the square of the changes, and the maximum change, respectively.) More formally, consider a M -dimensional societal decision problem in $X \subset \mathbb{R}^M$ and a population of voters V , where each voter $v \in V$ has bounded utility $fv(x) \in \mathbb{R}$, $\forall x \in X$. Then we consider the class of algorithms described in Algorithm 1. We study the algorithm class under two plausible models of how voters respond to query (1), which asks for the voter's favorite point in a local region.

- Model A: One possibility is that voters exactly perform the maximization asked of them, responding with their favorite point in the given L_q norm constraint set. In other words, they return a point $\arg \max_{x \in \{s : \|s - x_{t-1}\|_q \leq r_t\}} fv(x)$. Note that by definition of this movement, the algorithm is myopically incentive compatible: if a voter is the last voter and no projections are used, then truthfully performing this movement is the dominant strategy. In general, the mechanism is not globally incentive compatible, nor incentive compatible with projections onto the feasible region. Simple examples of manipulations in both instances exist.

- Model B: On the other hand, voters may not actually search within the constraint set to best improve the current point and then move in that direction to the boundary of the given constraint set. This model leads to a voter moving the current solution in the direction of the gradient of her utility function, returning a point $x_{t-1} + r_t \cdot g_t$ where $g_t \in \partial fv_t(x_{t-1})$. Note that $\partial f(x)$ denotes the set of subgradients of a function f at x , i.e. $g \in \partial f(x)$ if $\forall y, f(y) - f(x) \geq g^T (y - x)$.

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Point : Type u_2} (U : GKGMM19IterativeLocalVoting.ILVUtilityFormulaData Voter Point)
  (q : GKGMM19IterativeLocalVoting.SourceNorm) (center : Point) (r : ℝ) (voter : Voter) (response : Point),
  GKGMM19IterativeLocalVoting.modelAResponseFormulaData U q center r voter response →
  response ∈ GKGMM19IterativeLocalVoting.localNeighborhoodFormulaData U.geometry q center r ∧
  ∀ candidate ∈ GKGMM19IterativeLocalVoting.localNeighborhoodFormulaData U.geometry q center r,
  U.utility voter candidate ≤ U.utility voter response
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.modelA_response_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 29

Source locator: cited publication, lines 118-145

Verbatim source input

bounds on the sum of absolute values of the changes, the sum of the square of the changes, and the maximum change, respectively.) More formally, consider a M -dimensional societal decision problem in $X \subset \mathbb{R}^M$ and a population of voters V , where each voter $v \in V$ has bounded utility $f_v(x) \in \mathbb{R}$, $\forall x \in X$. Then we consider the class of algorithms described in Algorithm 1. We study the algorithm class under two plausible models of how voters respond to query (1), which asks for the voter's favorite point in a local region.

- Model A: One possibility is that voters exactly perform the maximization asked of them, responding with their favorite point in the given L_q norm constraint set. In other words, they return a point $\arg \max_{x \in \{s: \|s - x_{t-1}\|_q \leq r_t\}} f_v(x)$. Note that by definition of this movement, the algorithm is myopically incentive compatible: if a voter is the last voter and no projections are used, then truthfully performing this movement is the dominant strategy. In general, the mechanism is not globally incentive compatible, nor incentive compatible with projections onto the feasible region. Simple examples of manipulations in both instances exist.
- Model B: On the other hand, voters may not actually search within the constraint set to find their favorite point inside of it. Rather, a voter v may have an idea about how to best improve the current point and then move in that direction to the boundary of the given constraint set. This model leads to a voter moving the current solution in the direction of the gradient of her utility function, returning a point $x_{t-1} + r_t \frac{\nabla f_v(x_{t-1})}{\|\nabla f_v(x_{t-1})\|_q}$, for some $\nabla f_v(x_{t-1}) \in \partial f_v(x_{t-1})$. Note that $\partial f(x)$ denotes the set of subgradients of a function f at x , i.e. $g \in \partial f(x)$ if $\forall y, f(y) - f(x) \geq g^T (y - x)$.

Expanded PaperInterface specification

```

V {Voter : Type u_1} {Point : Type u_2} (U : GKGMM19IterativeLocalVoting.ILVUtilityFormulaData Voter Point)
  (q : GKGMM19IterativeLocalVoting.SourceNorm) (center : Point) (r : ℝ) (voter : Voter) (response : Point),
  GKGMM19IterativeLocalVoting.modelAResponseFormulaData U q center r voter response →
  response ∈ GKGMM19IterativeLocalVoting.localNeighborhoodFormulaData U.geometry q center r ∧
  IsMaxOn (U.utility voter) (GKGMM19IterativeLocalVoting.localNeighborhoodFormulaData U.geometry q center r)
  response
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.modelA_response_isMaxOn_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 30

Source locator: cited publication, lines 118-145

Verbatim source input

bounds on the sum of absolute values of the changes, the sum of the square of the changes, and the maximum change, respectively.)
More formally, consider a M -dimensional societal decision problem in $X \subset \mathbb{R}^M$ and a population of voters V , where each voter $v \in V$ has bounded utility $f_v(x) \in \mathbb{R}$, $\forall x \in X$. Then we consider the class of algorithms described in Algorithm 1. We study the algorithm class under two plausible models of how voters respond to query (1), which asks for the voter's favorite point in a local region.

- **Model A:**

One possibility is that voters exactly perform the maximization asked of them, responding with their favorite point in the given L_q norm constraint set. In other words, they return a point $\arg \max_{x \in \{s : \|s - x_{t-1}\|_q \leq r_t\}} f_v(x)$. Note that by definition of this movement, the algorithm is myopically incentive compatible: if a voter is the last voter and no projections are used, then truthfully performing this movement is the dominant strategy. In general, the mechanism is not globally incentive compatible, nor incentive compatible with projections onto the feasible region. Simple examples of manipulations in both instances exist.

- **Model B:** On the other hand, voters may not actually search within the constraint set to best improve the current point and then move in that direction to the boundary of the given constraint set. This model leads to a voter moving the current solution in the direction of the gradient of her utility function, returning a point $x_{t-1} + r_t \frac{\nabla f_v(x_{t-1})}{\|\nabla f_v(x_{t-1})\|_q}$, for some $\nabla f_v(x_{t-1}) \in \partial f_v(x_{t-1})$. Note that $\partial f(x)$ denotes the set of subgradients of a function f at x , i.e. $g \in \partial f(x)$ if $\forall y, f(y) - f(x) \geq g^T(y - x)$.

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Point : Type u_2} (U : GKGMM19IterativeLocalVoting.ILVUtilityFormulaData Voter Point)
  (p q : GKGMM19IterativeLocalVoting.SourceNorm) (ideal : Voter → Point) (center : Point) (r : ℝ) (voter : Voter)
  (response : Point),
  GKGMM19IterativeLocalVoting.isLpNormedUtilitiesFormulaData U ideal p →
  (GKGMM19IterativeLocalVoting.modelAResponseFormulaData U q center r voter response ↔
   response ∈ GKGMM19IterativeLocalVoting.localNeighborhoodFormulaData U.geometry q center r ∧
   IsMinOn (fun candidate => U.normDistance p candidate (ideal voter))
    (GKGMM19IterativeLocalVoting.localNeighborhoodFormulaData U.geometry q center r) response)
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.modelA_response_lp_normed_cost_minimizer_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 31

Source locator: cited publication, lines 118-145

Verbatim source input

bounds on the sum of absolute values of the changes, the sum of the square of the changes, and the maximum change, respectively.)
More formally, consider a M -dimensional societal decision problem in $X \subset \mathbb{R}^M$ and a population of voters V , where each voter $v \in V$ has bounded utility $f_v(x) \in \mathbb{R}$, $\forall x \in X$. Then we consider the class of algorithms described in Algorithm 1. We study the algorithm class under two plausible models of how voters respond to query (1), which asks for the voter's favorite point in a local region.

- **Model A:**

One possibility is that voters exactly perform the maximization asked of them, responding with their favorite point in the given L_q norm constraint set. In other words, they return a point $\arg \max_{x \in \{s : \|s - x_{t-1}\|_q \leq r_t\}} f_v(x)$. Note that by definition of this movement, the algorithm is myopically incentive compatible: if a voter is the last voter and no projections are used, then truthfully performing this movement is the dominant strategy. In general, the mechanism is not globally incentive compatible, nor incentive compatible with projections onto the feasible region. Simple examples of manipulations in both instances exist.

- **Model B:** On the other hand, voters may not actually search within the constraint set to find their favorite point inside of it. Rather, a voter v may have an idea about how to best improve the current point and then move in that direction to the boundary of the given constraint set. This model leads to a voter moving the current solution in the direction of the gradient of her utility function, returning a point $x_{t-1} + r_t \frac{\nabla f_v(x_{t-1})}{\|\nabla f_v(x_{t-1})\|_q}$, for some $\nabla f_v(x_{t-1}) \in \partial f_v(x_{t-1})$. Note that $\partial f(x)$ denotes the set of subgradients of a function f at x , i.e. $g \in \partial f(x)$ if $\forall y, f(y) - f(x) \geq g^\top (y - x)$.

Expanded PaperInterface specification

```
∀ {Coord : Type u_1} [inst : Fintype Coord] [inst_1 : Nonempty Coord] (q : GKGMM19IterativeLocalVoting.SourceNorm)
  (center : Coord → ℝ) (r : ℝ) (gradient_response : Coord → ℝ),
  GKGMM19IterativeLocalVoting.ModelBFiniteResponseAt q center r gradient_response ↔
    response = fun i => center i + r * (gradient i / GKGMM19IterativeLocalVoting.finiteCoordinateNorm q gradient)
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.modelB_finite_response_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 32

Source locator: cited publication, lines 118-145

Verbatim source input

bounds on the sum of absolute values of the changes, the sum of the square of the changes, and the maximum change, respectively.)
More formally, consider a M -dimensional societal decision problem in $X \subset \mathbb{R}^M$ and a population of voters V , where each voter $v \in V$ has bounded utility $fv(x) \in \mathbb{R}$, $\forall x \in X$. Then we consider the class of algorithms described in Algorithm 1. We study the algorithm class under two plausible models of how voters respond to query (1), which asks for the voter's favorite point in a local region.

- Model A:

One possibility is that voters exactly perform the maximization asked of them, responding with their favorite point in the given L_q norm constraint set. In other words, they return a point $\arg \max_{x \in \{s : \|s - x_{t-1}\|_q \leq r_t\}} fv_t(x)$. Note that by definition of this movement, the algorithm is myopically incentive compatible: if a voter is the last voter and no projections are used, then truthfully performing this movement is the dominant strategy. In general, the mechanism is not globally incentive compatible, nor incentive compatible with projections onto the feasible region. Simple examples of manipulations in both instances exist.

- Model B: On the other hand, voters may not actually search within the constraint set to best improve the current point and then move in that direction to the boundary of the given constraint set. This model leads to a voter moving the current solution in the direction of the gradient of her utility function, returning a point $x_{t-1} + r_t \cdot g_t$ where $g_t \in \partial fv_t(x_{t-1})$. Note that $\partial f(x)$ denotes the set of subgradients of a function f at x , i.e. $g \in \partial f(x)$ if $\forall y, f(y) - f(x) \geq g^T (y - x)$.

Expanded PaperInterface specification

```
∀ {Coord : Type u_1} [inst : Fintype Coord] [inst_1 : Nonempty Coord] {p q : ℝ},
  GKGM19IterativeLocalVoting.HolderDualFinite p q →
  ∀ {center ideal : Coord → ℝ},
    (∀ (i : Coord), center i ≠ ideal i) →
    ∀ (r : ℝ) (response : Coord → ℝ),
      GKGM19IterativeLocalVoting.ModelBFiniteResponseAt (GKGM19IterativeLocalVoting.SourceNorm.lp q) center r
        (fun i => -GKGM19IterativeLocalVoting.lpCostGradientCandidate p (fun j => center j - ideal j) i)
        response ↔
        response = fun i =>
          center i - r * GKGM19IterativeLocalVoting.lpCostGradientCandidate p (fun j => center j - ideal j) i
```

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.modelB_finite_response_neg_lp_cost_gradient_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 33

Source locator: cited publication, lines 480-535

Verbatim source input

sampling such a voter decreases fast enough).
Next, we introduce another general class of utility functions, which we call Weighted Euclidean utilities, for which one can obtain convergence to a unique solution.

Definition 2. Weighted Euclidean utilities. Let the solution space X be decomposable into K disjoint sub-spaces, so that $x = (x_1, \dots, x_K)$ for each $x \in X$ (where $\sum_{k=1}^K \dim(x_k) = M$). Suppose that the utility function of the voter v is

$$f_v(x) = - \sum_{k=1}^K w_k \frac{\|w_v\|^2}{\|x_k - x_k^v\|^2},$$

where w_v is a voter-specific weight vector, then the function is a Weighted Euclidean utility function. We further assume that $w_v \in \mathbb{R}^K$ and x_v are independently drawn for each voter v from a joint probability distribution with a bounded and measurable density function, with W nonempty, bounded, closed, and convex.

This utility function can be interpreted as follows: the decision-making problem is decomposable into K sub-problems, and each voter v has an ideal point x_k^v and a weight w_k^v for each sub-problem k , so that the voter's disutility for a solution is the weighted sum of the Euclidean distances to the ideal points in each sub-problems. Such utility functions may emerge in facility location problems, for example, where voters have preferences on

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[form-feed in source extraction]

Garg, Kamble, Goel, Marn, & Munagala

the locations of multiple facilities on a map. This utility form is also the one most closely related to the existing literature on Directional Equilibria and Quadratic Voting, in which preferences are linear. To recover the weighted linear preferences case, set $K = M$, with each sub-space of dimension 1. In this case, the following holds:

Proposition 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are Weighted Euclidean, and voters correctly respond to query (1) according to either Model A or Model B. Then, ILV with L2 neighborhoods converges with probability 1 to the societal optimal point.

The intuition for the result is as follows: as long as the neighborhood does not contain the ideal point of the sampled voter, the correct response to query (1) under weighted Euclidean preferences is to move the solution in the direction of the ideal point to the neighborhood boundary, which, as it turns out, is the same as the direction of the gradient. Thus with radius r_t , the effective movement is

Expanded PaperInterface specification

$\forall \{ \text{Voter} : \text{Type } u_1 \} \{ \text{Coord} : \text{Type } u_2 \} [\text{inst} : \text{Fintype Coord}] [\text{inst}_1 : \text{Nonempty Coord}]$
($E : \text{GKGMM19IterativeLocalVoting.ILVEnvironment Voter (Coord} \rightarrow \mathbb{R})$),
 $\text{GKGMM19IterativeLocalVoting.proposition1_weighted_euclidean_l2 } E = \text{GKGMM19IterativeLocalVoting.proposition1Statement } E$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.proposition1_weighted_euclidean_l2

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 34

Source locator: cited publication, lines 540-575

Verbatim source input

normalized (i.e. not divided by $\|w\|_2$), the algorithm would converge to the same point, as if utility functions were normalized.

3.2 Decomposable Utilities

Next consider the general class of decomposable utilities, motivated by the fact that the algorithm with L_∞ neighborhoods is of special interest since they are easy for humans to understand: one can change each dimension up to a certain amount, independent of the others.

Definition 3. Decomposable utilities. A voter utility function is decomposable if there exists concave functions f_m

v
for $m \in \{1 \dots M\}$ such that $f_v(x) = \sum_{m=1}^M f_m(x_m)$.

If the utility functions for the voters are decomposable, then we can show that our algorithm under L_∞ neighborhoods converges to the vector of medians of voters' ideal points on each dimension. Suppose that h_m
 X is the marginal density function of the random variable x_m
 v , and let $\tilde{x}_m$ be the set of medians of x_m
 v . (By set of medians, we mean the set of points such that, on each dimension, the mass of voters with ideal points above and below.)

Proposition 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are decomposable, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_∞ neighborhoods converges with probability 1 to a point in the set of medians $\tilde{x}$.

Although simply eliciting each agent's optimal solution and computing the vector of median allocations on each dimension is a viable approach in the case of decomposable utilities, deciding an optimal allocation across multiple dimensions is a more challenging cognitive task than deciding whether one wants to increase or decrease each dimension relative to the current solution (see Section 5.2.3 for experimental evidence). In fact, in this

Expanded PaperInterface specification

$\forall \{ \text{Voter Coord} : \text{Type} \} [\text{inst} : \text{Fintype Coord}] [\text{inst}_1 : \text{Nonempty Coord}]$
 $(E : \text{GKGMM19IterativeLocalVoting.ILVEnvironment Voter (Coord} \rightarrow \mathbb{R})),$
 $\text{GKGMM19IterativeLocalVoting.proposition2_decomposable_linf_medians } E =$
 $\text{GKGMM19IterativeLocalVoting.proposition2Statement } E$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.proposition2_decomposable_linf_medians

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 35

Source locator: cited publication, lines 1944-1965

Verbatim source input

Theorem 4. (Nemirovski et al., 2009; Strassen, 1965) Let $\theta \in \Theta$ be a random vector with distribution P . Let $\tilde{f}(x) = E[f(x, \theta)] =$

$\mathbb{E}[f(x, \theta)]$.

$\mathbb{E}[f(x, \theta)]dP(\theta)$, for $x \in X$, a non-empty bounded closed convex set, and assume the expectation is well-defined and $\mathbb{E}[f(x, \theta)]$ is finite valued. $\hookrightarrow$ Suppose that $f(\cdot, \theta)$, $\theta \in \Theta$ is convex and $\tilde{f}(\cdot)$ is continuous and $\mathbb{E}[f(x, \theta)]$ is finite valued in a neighborhood of point x . For each θ , choose any $g(x, \theta) \in \partial f(x, \theta)$. Then, there exists $\tilde{g}(x) \in \partial \tilde{f}(x)$ s.t. $\tilde{g}(x) = \mathbb{E}[g(x, \theta)]$.

This theorem says that the expected value of the sub-gradient of the utility at any point x across voters is a subgradient of the societal utility at x , irrespective of how the voters choose the subgradient when there are multiple subgradients, i.e., when the utility function is not differentiable. This key result allows us to use the subgradient of utility function of a sampled voter as an unbiased estimate of the societal subgradient. Now, consider a convex function f on a non-empty bounded closed convex set $X \subset \mathbb{R}^M$, and use $[\cdot]_X$ to designate the projection operator. Starting with some $x_0 \in X$, consider the SSGM update rule $x_t = [x_{t-1} - \eta_t(\tilde{g}_t + z_t + b_t)]_X$, where z_t is a zero-mean random variable and b_t is a constant, and $\tilde{g}_t \in \partial f(x_t)$. Let $E_t[\cdot]$ be the conditional expectation given F_t , the σ -algebra generated by $x_0, x_1, \dots, x_t$. Then we have the following convergence result.

Expanded PaperInterface specification

```
∀ {Theta : Type u_1} {Coord : Type u_2} [inst : MeasurableSpace Theta] [inst_1 : Fintype Coord]
(mu : MeasureTheory.Measure Theta) [MeasureTheory.IsProbabilityMeasure mu] (solutionSpace : Set (Coord → ℝ))
(sampleCost : Theta → (Coord → ℝ) → ℝ) (x : Coord → ℝ) (sampleGradient : Theta → Coord → ℝ),
solutionSpace.Nonempty →
Bornology.IsBounded solutionSpace →
IsClosed solutionSpace →
Convex ℝ solutionSpace →
x ∈ solutionSpace →
(∀ y ∈ solutionSpace, MeasureTheory.Integrable (fun theta => sampleCost theta y) mu) →
(∀ (i : Coord), MeasureTheory.Integrable (fun theta => sampleGradient theta i) mu) →
(∀ (theta : Theta), ConvexOn ℝ solutionSpace (sampleCost theta)) →
ContinuousAt (fun y => ∫ (theta : Theta), sampleCost theta y ∂mu) x →
(∀ (theta : Theta),
  KGMM19IterativeLocalVoting.FiniteSubgradientWithinAt (sampleCost theta) solutionSpace x
    (sampleGradient theta)) →
KGMM19IterativeLocalVoting.ExpectedSubgradientTheoremStatement mu solutionSpace sampleCost x
  sampleGradient
```

Lean proof endpoint (built separately)

KGMM19IterativeLocalVoting.appendix_theorem4_expected_subgradient_boundary

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 36

Source locator: cited publication, lines 1971-2005

Verbatim source input

Theorem 5. (Jiang & Walrand, 2010) Consider the above update rule. If

$f(\cdot)$ has a unique minimizer $x^* \in X$

$r_t > 0$,

$\emptyset$

t
 $r_t = \infty$,

$\emptyset$

t
 r_t^2
 $t < \infty$

$\exists C_1 \in \mathbb{R} < \infty$ s.t. $\|\partial f(x)\|^2 \leq C_1, \forall x \in X$

$\exists C_2 \in \mathbb{R} < \infty$ s.t. $\mathbb{E}[\|z_t\|^2] \leq C_2, \forall t$

$\exists C_3 \in \mathbb{R} < \infty$ s.t. $\|b_t\|^2 \leq C_3, \forall t$

$\emptyset$

t
 $r_t \|b_t\| < \infty$ w.p. 1

Then $x_t \rightarrow x^*$ w.p. 1 as $t \rightarrow \infty$.

Note: Jiang and Walrand (2010) prove the result for gradients, though the same proof follows for subgradients. Only the inequality $[x^* - x_t]^T g_t \leq f(x^*) - f(x_t)$ for gradient g_t at iteration t is used, which holds for subgradients. Boyd and Mutapcic (2006) provide a general discussion of subgradient methods, along with similar results. Shor (1998), in Theorem 46, provide a convergence proof for the stochastic subgradient method without projections and the extra noise terms.

C.2 Mapping ILV to SSGM

Expanded PaperInterface specification

```
∀ {Ω : Type u_1} {Coord : Type u_2} {mΩ : MeasurableSpace Ω} [inst : Fintype Coord]
[inst_1 : Nonempty Coord] (μ : MeasureTheory.Measure Ω) [inst_2 : MeasureTheory.IsProbabilityMeasure μ]
(filtration : MeasureTheory.Filtration ℕ mΩ) (solutionSpace : Set (Coord → ℝ)) (objective : (Coord → ℝ) → ℝ)
(project : (Coord → ℝ) → Coord → ℝ) (trajectory meanSubgradient noise : ℕ → Ω → Coord → ℝ) (bias : ℕ → Coord → ℝ)
(radius : ℕ → ℝ) (xstar : Coord → ℝ),
GKGM19IterativeLocalVoting.AppendixTheorem5Statement μ filtration solutionSpace objective project trajectory
meanSubgradient noise bias radius xstar
```

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.appendix_theorem5_ssgm_convergence_boundary

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 37

Source locator: cited publication, lines 645-680

Verbatim source input

325

[form-feed in source extraction]

Garg, Kamble, Goel, Marn, & Munagala

Rather, we consider an unconstrained budget allocation problem, one in which a voter's utility includes a term for the budget de[U+001C control character]cit. Let $E \subseteq \{1 \dots M\}$, $I = \{1 \dots M\} \setminus E$ be the expenditure and income items, respectively. Then the general budget utility function is $f_v(x) = g_v(x) - d(\sum_{i \in I} x_i)$, where d is an increasing function on the de[U+001C control character]cit. For example, suppose a voter's disutility was proportional to the square of the budget de[U+001C control character]cit (she especially dislikes large budget de[U+001C control character]cits); then, this term adds $\hookrightarrow$ complex dependencies between the budget items. In general, nothing is known about convergence of Algorithm 1 with such utilities, as the de[U+001C control character]cit term may add complex dependencies between the dimensions. However, if the voter utility functions are decomposable across the dimensions and L -neighborhoods used, then the results of Section 3.2 can be applied. We propose the following class of decomposable utility functions for the budgeting problem, achieved by assuming that the cost for the de[U+001C control character]cit is linear, and call the class [U+0010 control character]decomposable with a linear cost for de[U+001C control character]cit," or DLCD.

Definition 4. Let $f_v(x)$ be DLCD if

$$f_v(x) = \sum_{m=1}^M f_m(v(x_m)) - w_v$$

Expanded PaperInterface specification

```
∀ {Dim : Type u_1} [inst : Fintype Dim] (utility : (Dim → ℝ) → ℝ) (isExpense : Dim → Bool)
  (componentUtility : Dim → ℝ → ℝ) (deficitWeight : ℝ),
  GKGMM19IterativeLocalVoting.paper_definition4_dlcd_formula utility isExpense componentUtility deficitWeight =
    (0 ≤ deficitWeight ∧
     (∀ (m : Dim), ConcaveOn ℝ Set.univ (componentUtility m)) ∧
     ∀ (x : Dim → ℝ),
       utility x = GKGMM19IterativeLocalVoting.dlcdBudgetUtility isExpense componentUtility deficitWeight x)
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.paper_definition4_dlcd_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 38

Source locator: cited publication, lines 645-680

Verbatim source input

325

[form-feed in source extraction]

Garg, Kamble, Goel, Marn, & Munagala

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$$f_v(x) = g_v(x) - d(\sum_{i \in I} x_i)$$
 where d is an increasing function on the de[U+001C control character]cit. For example, suppose a voter's disutility was proportional to the square of the budget de[U+001C control character]cit (she especially dislikes large budget de[U+001C control character]cits); then, this term adds $\hookrightarrow$ complex dependencies between the budget items. In general, nothing is known about convergence of Algorithm 1 with such utilities, as the de[U+001C control character]cit term may add complex dependencies between the dimensions. However, if the voter utility functions are decomposable across the dimensions and L -neighborhoods used, then the results of Section 3.2 can be applied. We propose the following class of decomposable utility functions for the budgeting problem, achieved by assuming that the cost for the de[U+001C control character]cit is linear, and call the class [U+0010 control character]decomposable with a linear cost for de[U+001C control character]cit," or DLCD.

Definition 4. Let $f_v(x)$ be DLCD if

$$f_v(x) = \sum_{m=1}^M \sum_{i \in I} v_i(x_m) - w_v \sum_{i \in E} x_i$$

Expanded PaperInterface specification

$\forall \{Dim : \text{Type } u_1\} [inst : \text{Fintype } Dim] (isExpense : Dim \rightarrow Bool) (componentUtility : Dim \rightarrow \mathbb{R} \rightarrow \mathbb{R}) (deficitWeight : \mathbb{R}) (x : Dim \rightarrow \mathbb{R}),$
GKGMM19IterativeLocalVoting.dlcdBudgetUtility isExpense componentUtility deficitWeight x =
 $\sum m, componentUtility\ m\ (x\ m) - deficitWeight * ((\sum m, if\ isExpense\ m = true\ then\ x\ m\ else\ 0) - \sum m, if\ isExpense\ m = true\ then\ 0\ else\ x\ m)$

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.dlcdBudgetUtility

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 39

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. L_p normed utilities. The voter utility function is L_p normed if $f_v(x) = -||x - x_v||_p$, $\forall x \in X$.

Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2)$, $(1, \infty)$, or $(\infty, 1)$.

322

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Coord : Type u_2} [inst : Fintype Coord] [inst_1 : Nonempty Coord]
  (E : GKGMM19IterativeLocalVoting.ILVEnvironment Voter (Coord → ℝ)),
  GKGMM19IterativeLocalVoting.theorem1_lp_normed_dual_cases E = GKGMM19IterativeLocalVoting.theorem1Statement E
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.theorem1_lp_normed_dual_cases

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 40

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. L_p normed utilities. The voter utility function is L_p normed if $fv(x) = -||x - xv||_p, \forall x \in X$.

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Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2), (1, \infty),$ or $(\infty, 1)$.

322

Expanded PaperInterface specification

GKGM19IterativeLocalVoting.Theorem1NormPair GKGM19IterativeLocalVoting.SourceNorm.l1
GKGM19IterativeLocalVoting.SourceNorm.linf

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.theorem1_norm_pair_l1_linf

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 41

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

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Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2), (1, \infty)$, or $(\infty, 1)$.

322

Expanded PaperInterface specification

GKGMM19IterativeLocalVoting.Theorem1NormPair GKGMM19IterativeLocalVoting.SourceNorm.l2
GKGMM19IterativeLocalVoting.SourceNorm.l2

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.theorem1_norm_pair_l2_l2

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 42

Source locator: cited publication, lines 430-450

Verbatim source input

3.1 Spatial Utilities

Here we consider spatial utility functions, where the utilities of each voters can be expressed in the form of some kind of spatial distance from their ideal solutions. First, we consider the following kind of utilities.

Definition 1. L_p normed utilities. The voter utility function is L_p normed if $fv(x) = -||x - xv||_p, \forall x \in X$.

Under such utilities, for $p = 1, 2$ and ∞ , restricting voters to a ball in the dual norm leads to convergence to the societal optimum.

Theorem 1. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to either Model A or Model B. Then, ILV with L_q neighborhoods converges to the societal optimal point w.p. 1 when $(p, q) = (2, 2), (1, \infty)$, or $(\infty, 1)$.

322

Expanded PaperInterface specification

GKGMM19IterativeLocalVoting.Theorem1NormPair GKGMM19IterativeLocalVoting.SourceNorm.linfity
GKGMM19IterativeLocalVoting.SourceNorm.l1

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.theorem1_norm_pair_linf_l1

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 43

Source locator: cited publication, lines 460-480

Verbatim source input

rule is not equivalent to SSGM in general, and we leave the convergence to the societal optimum for general (p, q) as an open question. Further, note that even if each voter could scale their utility function arbitrarily, the algorithm would converge to the same point. However, the general result does hold for general dual norms (p, q) if one assumes the alternative behavior model.

Theorem 2. Suppose that conditions C1, C2, and C3 are satisfied, the voter utilities are L_p normed, and voters respond to query (1) according to Model B. Then, ILV with L_q

neighborhoods converges to the societal optimal point w.p. 1 for any $p > 0$ and $q > 0$ such that $1/p + 1/q = 1$.

The proof is contained in the appendix. It uses the following property of L_p normed utilities: the L_q norm of the gradient of these utilities at any point other than the ideal point is constant. This fact, along with the voter behavior model, allows the algorithm to implicitly capture the magnitude of the gradient of the utilities, and thus a direct mapping to SSGM is obtained.

Note that the above result holds even if we assume that a voter moves to her ideal point x_v in case it falls within the neighborhood (since, as explained earlier, the probability of sampling such a voter decreases fast enough).

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Coord : Type u_2} [inst : Fintype Coord] [inst_1 : Nonempty Coord]
  (E : GKGMM19IterativeLocalVoting.ILVEnvironment Voter (Coord → ℝ)),
  GKGMM19IterativeLocalVoting.theorem2_modelB_holder_dual_norms E = GKGMM19IterativeLocalVoting.theorem2Statement E
```

Lean proof endpoint (built separately)

GKGMM19IterativeLocalVoting.theorem2_modelB_holder_dual_norms

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 44

Source locator: cited publication, lines 590-620

Verbatim source input

ited to simulations. We show that with the radius decreasing as $O(1/t)$, the algorithm indeed finds directional equilibria in the following sense: if under a few conditions a trajectory of the algorithm converges to a point, then that point is a directional equilibrium.

Theorem 3. Suppose that C_1 , C_2 , and C_3 are satisfied, and let $G(x) = \nabla f_v(x)$.

Suppose, $G(x)$ is uniformly continuous, L_2 movement norm constraints are used, and voters move according to Model B. If a trajectory $\{x_t\}_{t=1}^\infty$ of the algorithm converges to x^* , i.e. $\lim_{t \rightarrow \infty} x_t = x^*$, then x^* is a directional equilibrium, i.e. $G(x^*) = 0$.

The proof is in the appendix. It relies heavily on the continuity assumption: if a point x is not a directional equilibrium, then the algorithm with step sizes $O(1/t)$ will with probability 1 leave any small region surrounding x : the net drift of the voter movements is away from the region. We note that the necessary assumptions hold for all utility functions for which convergence holds, using the L_2 norm algorithm (e.g. weighted Euclidean utilities). It is further possible to characterize other utility functions for which the equivalence holds: with appropriate conditions on the distribution of voters and how they differ among voters, the conditions on G can be met.

We further conjecture that even under voter Model A, if Algorithm 2 converges, the fixed point is a Directional Equilibrium. Note that as $t \rightarrow \infty$, $f_v(y)$ can be linearly approximated

Expanded PaperInterface specification

```
∀ {Voter : Type u_1} {Coord : Type u_2} [inst : Fintype Voter] [inst_1 : MeasurableSpace Voter]
  [inst_2 : MeasurableSingletonClass Voter] [inst_3 : Fintype Coord] [inst_4 : Nonempty Coord]
  (E : GKGM19IterativeLocalVoting.ILVEnvironment Voter (Coord → ℝ))
  (M : GKGM19IterativeLocalVoting.FiniteCoordinateTheorem3SampledProjectedSourceSemantics E),
  E.solutionSpace = Set.univ → GKGM19IterativeLocalVoting.theorem3Statement E
```

Lean proof endpoint (built separately)

GKGM19IterativeLocalVoting.theorem3_statement_of_full_sampled_projected_source_semantics_univ

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation